

Reconstructing Audio from Silent Video

A Three-Path System for Speech, Foley and Surface-Vibration Recovery on Consumer Hardware

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Abstract—Video recorded without usable sound is common, and the information needed to reconstruct plausible audio is often still present in the pixels. This paper describes a system that recovers audio from silent video along three independent paths: visual speech recognition followed by timed speech synthesis; action recognition followed by generated Foley that is aligned to measured visual events; and phase-based recovery of surface vibrations. All three run locally on a single Apple M4 laptop with 17 GB of unified memory and read the video stream only. The central engineering contribution is in the Foley path. Action recognition returns intervals such as “walking, 1.5–2.5 s”, but a footstep is audible at an instant rather than across a span, so the system performs an independent frame-level motion analysis to locate the exact frames of contact and anchors each generated sound to those instants using true envelope attack times. Generated audio is not trusted: every asset is measured before mixing and must clear six quality gates. Across the 54 assets generated during development, the gate rejected 16 (29.6 %), of which 11 would have required more than +25 dB of automatic make-up gain. On the reference clip the worst synchronisation error measured on the rendered audio is 20 ms, below half a frame interval at 24 fps. Two ablations show that the aggregate quality score is insensitive to generation settings that audibly degrade output: reducing the denoised latent from 30 s to 10 s cut the number of detectable transients in a walking asset from 16 to 8 and raised inter-step irregularity by a factor of 5.8, while the measured dynamic range *improved* and the aggregate score remained far above the acceptance threshold. A controlled synthetic experiment characterises the vibration path, which recovers a known sub-pixel oscillation exactly down to a displacement of 0.02 px and aliases about the Nyquist frequency exactly as theory predicts. The system is a single-machine demonstrator, and the limits of each path are reported rather than smoothed over.

Index Terms— Foley synthesis, video-to-audio generation, visual speech recognition, audio-visual synchronisation, diffusion transformer, visual microphone, quality gating.

I. INTRODUCTION

A large amount of recorded video carries no usable audio. Surveillance cameras often record picture only, archival footage may have lost or never had a sound track, and consumer recordings are routinely spoiled by wind, handling noise, or a microphone that was muted. In each case the visual record still contains substantial information about what was happening acoustically: a person's lips move in a way that constrains what they said, a mug meeting a table implies a particular contact sound at a particular instant, and the surfaces in the frame vibrate in response to the sound field around them.

This paper describes a system built around that observation. It takes a silent video and produces a video with a generated audio track, using three independent paths that address three different kinds of acoustic information:

- **S1 — Lip reading and speech generation.** A visual speech recognition model reads lip motion, and the recognised sentence is synthesised and placed against word onsets derived from the video itself.
- **S2 — Action recognition and Foley generation.** A vision-language model labels physical actions over time, a text-to-audio model generates matching Foley, and each generated sound is aligned to a visually measured contact instant.

- **S3 — Acoustic Eye.** A phase-based visual microphone recovers a signal from the sub-pixel vibration of surfaces in the frame.

The three paths are complementary rather than competing. S1 addresses speech, S2 addresses physical interaction, and S3 addresses the acoustic field itself. They share an input contract — the video stream only is decoded, and any audio already present in the container is never read — and an output contract, a playable file whose picture is stream-copied without re-encoding.

The engineering depth of the work is concentrated in S2, and so is the emphasis of this paper. Two problems there proved harder than expected and shaped the design.

The first is *where* to put a generated sound. An action recogniser reports that walking occupies 1.5–2.5 s, but that interval is not when a footstep is audible; the audible events are the individual foot plants, which are instants. Placing a two-second walking clip across a two-second label produces audio that is present but visibly wrong. The system therefore performs an independent frame-level motion analysis whose only job is to find those instants, and anchors audio to them.

The second is that generative audio models fail in ways that conventional level-setting makes worse. A text-to-audio model occasionally returns a near-silent, near-constant output containing no usable signal. An active-RMS leveller will faithfully attempt to raise such a file to its target level,

applying tens of decibels of gain and converting quantisation noise into audible hiss. The system therefore measures every generated asset before it can enter the mix, and refuses material that cannot be used — leaving the corresponding interval silent and reporting why, rather than substituting something else.

A. Contributions

1. A working, locally-run system that converts silent video to video with audio along three distinct paths, with all model inference isolated in separate processes to satisfy conflicting dependency and memory constraints on a 17 GB machine.
2. A visual event localisation and alignment method that anchors generated Foley to measured contact instants using true envelope attack times rather than onset-strength peaks, with the resulting error measured on the rendered audio rather than asserted from the plan.
3. A multi-criteria quality gate for generated audio, evaluated by ablation over the full corpus of 54 assets produced during development.
4. Empirical evidence that aggregate audio quality scores are insensitive to generation settings that audibly degrade output, and that a structural measure — transient count and rhythmic regularity — detects what the scores miss.
5. A controlled characterisation of the visual microphone path establishing its sub-pixel detection floor and confirming its Nyquist behaviour.

What has not been established. No word error rate is reported for S1, because ground-truth transcripts for the test recordings were not retained. Listening judgements were made by a single assessor, the developer, and are reported as such. S3 has been validated on synthetic stimuli with known ground truth, and has *not* been shown to recover real sound from real footage on the hardware available.

II. RELATED WORK

A. Sound from silent video

Owens *et al.* [1] framed sound prediction from silent video as a way of probing physical scene understanding, training a recurrent network to predict sound features from video of a drumstick striking and scratching materials, then synthesising a waveform by example-based retrieval. The setting was deliberately constrained — one implement, controlled interactions — and the work established that visual evidence of contact is sufficient to constrain the resulting sound.

Recent video-to-audio systems condition a generative model directly on video. Diff-Foley [2] uses contrastive audio-visual pretraining to learn temporally aligned features and trains a latent diffusion model on spectrogram latents conditioned on them, reporting improved synchronisation over earlier methods. FoleyCrafter [3] attaches a semantic adapter and a temporal controller, including an onset detector, to a frozen text-to-audio backbone, and supports text prompts alongside the video condition. MMAudio [4] trains jointly on

video-audio and larger-scale text-audio data with a flow matching objective and a frame-level synchronisation module, reporting state-of-the-art results among public models at 157 M parameters.

Position relative to that work. Those systems learn the mapping from video to audio end-to-end, including timing. The system described here does not: it uses a text-conditioned audio model with *no video input at all*, and obtains timing from an explicit, measurable analysis of frame motion. This is a weaker form of audio-visual modelling and it will not generalise as broadly. It was adopted because it runs within the available memory budget, because its failures are inspectable, and — as Section III documents — because the video-conditioned models available at the time did not produce usable output for the target material on this hardware.

B. Text-to-audio generation

AudioLDM 2 [5] proposes a shared representation across speech, music and sound effects, generating through a latent diffusion model conditioned on a self-supervised audio representation. Stable Audio Open [6] is an open-weights latent diffusion model trained on Creative Commons material, producing 44.1 kHz stereo. MOSS-SoundEffect v2.0 [7], used here, is a 1.3 B-parameter Diffusion Transformer trained with a flow matching objective, with a Qwen3-1.7B text encoder and a DAC [8] variational autoencoder, producing 48 kHz audio of up to 30 s under an Apache-2.0 licence. The architecture follows the Diffusion Transformer formulation of Peebles and Xie [9] with the flow matching training objective of Lipman *et al.* [10].

C. Visual speech recognition

Afouras *et al.* [11] treated lip reading as open-vocabulary sentence recognition in unconstrained video and released the LRS2-BBC dataset. AV-HuBERT [12] learns audio-visual speech representations by masked prediction of iteratively refined multimodal cluster assignments, substantially reducing the labelled data required. Auto-AVSR [13], used in S1, increases effective training data by automatically transcribing unlabelled audio-visual corpora, and releases visual-only checkpoints built on a Conformer encoder [14] with a Transformer decoder. Word timings in S1 are obtained by forced alignment against the model's CTC [15] head.

D. Recovering sound from vibration

Davis *et al.* [16] showed that sound-induced vibrations of everyday objects can be recovered from high-speed video, reconstructing intelligible speech and music from footage of a chip bag and a potted plant. The method decomposes each frame with a complex steerable pyramid [18] and tracks local phase, following the phase-based motion analysis of Wadhwa *et al.* [17]. The dominant practical constraint is sampling: the recoverable bandwidth is bounded by half the frame rate, which is why the original work used cameras running at thousands of frames per second. S3 implements this method and, as Section IX reports, is bounded by exactly that constraint on ordinary footage.

E. Evaluation of generated audio

Fréchet Audio Distance [19] adapts the Fréchet Inception Distance to audio and is the standard aggregate metric for generative audio systems. It compares distributions over a corpus and is not applicable to the per-asset accept/reject decision required here, where a single generated file must be judged usable or not before it enters a mix. The gate described in Section VI is a different instrument for a different purpose, and is not proposed as a replacement for corpus-level metrics.

III. EXISTING APPROACHES EVALUATED AND REJECTED

Model selection was carried out empirically on the target hardware rather than from published benchmarks, because the binding constraints — 17 GB of unified memory, Metal Performance Shaders (MPS) rather than CUDA, and no cloud inference — are not the constraints those benchmarks were produced under. This section records what was tried and why it was set aside, because the negative results motivate the final design.

A. Action recognition

Two closed-vocabulary recognisers were tested before the vision-language model was adopted. A VideoMAE [20] checkpoint fine-tuned on Kinetics returned *shredding paper* with confidence 0.550 for a clip of a person handling a cup, with no relevant label in the top ten. An X-CLIP [21] temporal test returned *pouring liquid* at confidence 0.21–0.38 across every window of the same clip, collapsing to a single segment. The failure is structural rather than incidental: Kinetics-style label sets are organised around human activities at the scale of a whole clip, not around the object-contact events that generate Foley. Qwen2.5-VL-3B-Instruct [22] was adopted instead because it emits free text, so the vocabulary is not fixed in advance, and because a 3 B model in bfloat16 fits the memory budget.

B. Sound generation

TABLE I. TEXT-TO-AUDIO AND VIDEO-TO-AUDIO MODELS EVALUATED AND REJECTED, WITH THE MEASURED FAILURE MODE ON A COMMON DRINKING-FOLEY TEST CASE.

Model	Output	Failure mode
Stable Audio Open 1.0	44.1 kHz, 4.5 s	96.2 % silence, 2 clicks
Stable Audio Open Small	44.1 kHz, 3.0 s	95.6 % silence
AudioLDM 2	16 kHz, 10 s	91.9 % silence, 30 ms events
FoleyCrafter	16 kHz, 3.0 s	continuous noise bed
MMAudio v1 / v2	44.1 kHz	86 % / 96 % silence; wrong action

Three of the five failed with a strikingly consistent signature: over 91 % digital silence punctuated by isolated 30–40 ms impulses. Before attributing this to the models, we note that every one of those runs requested 2.0–4.5 s of audio from models trained to generate at 10–47 s. Requesting output far below the training duration is a plausible cause of out-of-regime behaviour, and the conclusion drawn was procedural rather than about model identity: generate at the model's native duration and crop afterwards. This is what the final system does, generating 10 s and using only the segment required.

MOSS-SoundEffect v2.0 was selected on four grounds: 48 kHz output, which matters for the high-frequency content of wet mouth sounds and ceramic contacts; duration control up to 30 s, which addresses the failure signature above; an explicitly documented human-action Foley category; and an Apache-2.0 licence, where MMAudio, FoleyCrafter and AudioGen are all non-commercial.

C. Direct comparison: MOSS versus Stable Audio Open

Stable Audio Open was retained as a switchable backend and run through the identical pipeline, which permits a controlled comparison. The discriminating measurement is the harmonic ratio. Foley is inharmonic — a mug meeting a table is a broadband transient — so a near-pure tone indicates failure regardless of how good other metrics look.

TABLE II. QUALITY SCORE AND HARMONIC RATIO FOR FOUR FOLEY CLASSES GENERATED THROUGH THE IDENTICAL PIPELINE BY BOTH BACKENDS. LOWER HARMONIC RATIO IS BETTER.

Class	MOSS score	MOSS harmonic	SA score	SA harmonic
Walking	97.1	0.00	92.7	0.03
Drinking	70.9	0.06	75.6	0.09
Cup pickup	85.8	0.00	53.1	0.88
Cup placement	49.8	0.02	53.4	0.87

The two backends are comparable on walking and drinking, and Stable Audio scores marginally higher on drinking and cup placement. On object contacts, however, Stable Audio produced musical tones: one output for “cup placed on a table” was a 346 Hz sine wave with a harmonic ratio of 0.997 and a

spectral flatness of 0.00000. Judged on the aggregate score alone, that output would have been accepted. This observation is what motivated adding an independent pure-tone gate, and is revisited quantitatively in Section IX.

IV. SYSTEM DESIGN

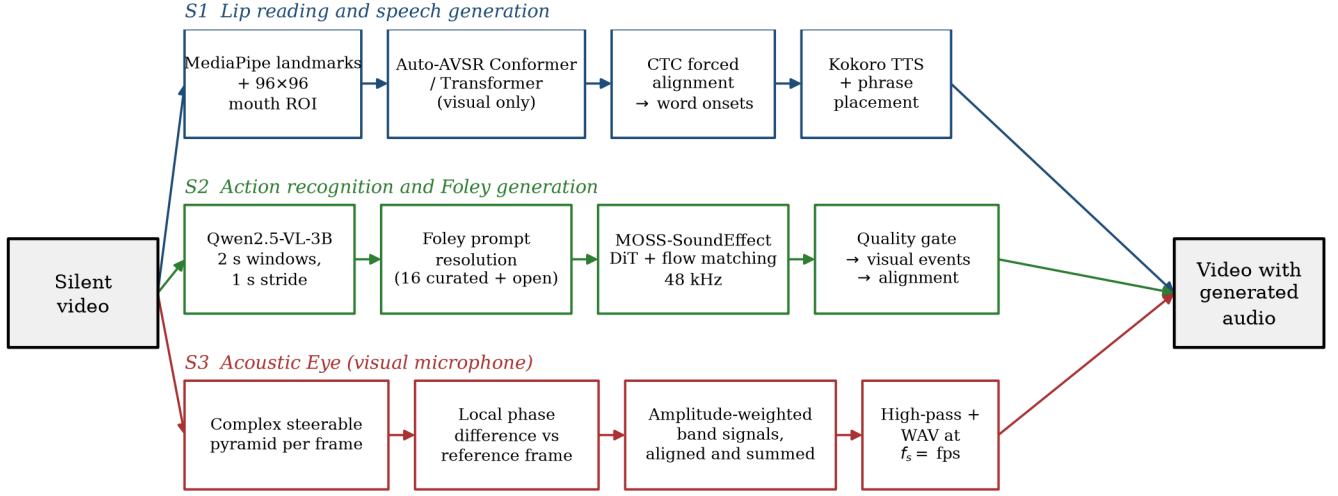


Fig. 1. System architecture. Three independent paths convert silent video to audio. All three decode the video stream only; audio present in the source container is never read. Each model runs in its own virtual environment as a subprocess, so no two models are ever resident in the same interpreter or at the same time.

A. Process isolation

The three paths impose mutually incompatible requirements: S1 needs Python 3.11 with torchvision 0.20.1 (whose video reader was removed in later releases) and `numpy<2` for MediaPipe [23], while S2 needs Python 3.12 with torch 2.9.1, and its text-to-speech component needs `numpy>=2`. These cannot coexist in one interpreter.

Models are therefore invoked as subprocesses in separate virtual environments, never imported into the long-lived API process. Three consequences follow, all deliberate. Memory is released unconditionally at process exit, which is more reliable than garbage collection combined with `torch.mps.empty_cache()`; this matters because Qwen2.5-VL peaks near 12 GB and MOSS near 12.1 GB on a 17 GB machine. Dependencies are isolated. And a model crash returns a non-zero exit code that is converted to a readable message rather than taking down the server.

B. The nine-stage pipeline (S2)

The S2 path runs as nine stages: upload, validation, action recognition, timeline resolution, Foley generation, Foley quality validation, visual synchronisation, audio mixing, and rendering. Each stage reports real state to a job store that is persisted to disk after every transition. Progress within action recognition is derived from a file the runner writes per window, so the reported percentage tracks completed work rather than elapsed time.

Validation rejects files with no video stream, duration outside 0.4–60 s, size above 200 MB, or a disallowed extension. A video that already carries an audio track is accepted with a warning; the stream is never decoded.

C. Action timeline

Qwen2.5-VL is run over 2.0 s windows at 1.0 s stride, eight frames per window resized to 448×252, in bfloat16 on MPS. Each window is described independently and the response is parsed into an action phrase and its verb head. Windows sharing a head are merged into spans.

Because the stride is half the window, raw spans overlap by one stride: for the reference clip, *pick up cup* spans 2.0–6.0 s and *drink from cup* spans 5.0–9.0 s, sharing a second. An instant belonging to two actions is ambiguous for audio placement, so spans are resolved to a non-overlapping timeline by midpoint boundary resolution. Segments supported by few windows are flagged *suspect* rather than deleted, and surface in the interface as medium confidence.

D. Foley class resolution

A resolved action phrase is mapped to a Foley specification comprising a generation prompt, a negative prompt, a synchronisation strategy, a frame region in which to measure motion, an asset selection rule, and a target level. Resolution proceeds in tiers: 16 hand-tuned curated classes are matched first, then a set of deliberately silent actions (standing, reaching, holding), then a verb-plus-object fallback to generic placement, pickup or press classes.

The ordering is load-bearing. An earlier version matched on bare keywords such as `place` and `pick`, which caused “place spoon on table” to inherit ceramic-mug Foley. Every curated keyword now names its object, and generic classes are tried only after specific ones. That precision opened a different hole: “place bread in toaster” is unmistakably a placement but contains none of the literal surfaces the generic keywords required, so it resolved to nothing and the job failed before reaching the audio model. The verb fallback closes that hole

while running last, so specific classes and deliberate silences still win.

A final open-vocabulary tier synthesises a specification for any phrase not otherwise matched. It classifies the phrase's verb into one of nine archetypes (locomotion, impact, placement, pickup, mechanism, friction, liquid, oral, ambient), each of which determines the synchronisation strategy, region, selection rule and level, and composes a prompt from the archetype's acoustic description and the literal action phrase. An unrecognised verb defaults to a continuous ambient texture rather than a sharp transient, because a misplaced texture is far less jarring than a misplaced impact. Synthesis is deterministic, so the same phrase always produces the same prompt and the content-addressed cache remains meaningful; keys are formed from archetype and object, so “kick the ball” and “kicking a football” share one generation.

E. Foley generation

Generation runs in three memory-separated phases so the components are never co-resident: the Qwen3 text encoder (3.44 GB), the Diffusion Transformer (2.85 GB), and the DAC decoder (0.74 GB). Parameters are cast to bfloat16 but buffers are left untouched, because the Transformer carries complex rotary position tables that a blanket cast would destroy. One embedding function is redirected to CPU because it computes in float64, which MPS does not support. Both adaptations live in a wrapper; the upstream repository is unmodified, which is verified by requiring `git status --porcelain` to return empty.

Assets are cached by a SHA-256 over the action key, both prompts, seed, step count, guidance scale, sigma shift, duration and sample rate, truncated to 16 hex characters. Numeric values are normalised before hashing — without this, a client sending `duration: 10` and a default of `10.0` produce different keys for byte-identical audio.

V. FOLEY QUALITY VALIDATION

Every generated asset is measured in its raw state, before any gain or normalisation, and must clear six gates to enter the mix:

1. effective bits ≥ 9.0 , where effective bits is $16 + \log_2(\text{peak})$, rejecting material whose peak is below about -42 dBFS;
2. dynamic range ≥ 6 dB, measured over signal-bearing frames only;
3. not a sustained tone: harmonic ratio > 0.80 combined with dynamic range < 10 dB;
4. not a pure tone: harmonic ratio > 0.90 combined with spectral flatness < 0.01 ;
5. required make-up gain $\leq +25$ dB;
6. integrity: finite samples, correct sample rate, duration ≥ 0.05 s.

Two of these thresholds were set by specific observed failures. Gate 4 exists because gate 3 alone missed the 346 Hz sine

described in Section III: that file had 32 dB of dynamic range, so the combined tonality test did not fire. A tone with a natural envelope is still a tone, and inharmonicity is disqualifying on its own.

The dynamic-range measurement in gate 2 requires care. Computing it over all frames including exact digital silence sends the fifth percentile to the numeric floor and returns values of 183–201 dB, which is physically impossible for 16-bit audio and scores full marks. A file that is 90 % silence with a few clicks would pass as excellent. The measurement is therefore restricted to frames above -70 dB relative to the frame maximum, and capped at 96 dB.

Passing candidates are ranked by a score in $[0, 100]$ combining dynamic range (40 points), signal level (25), gain headroom (20) and non-tonality (15). Up to three candidates are generated with successive seeds, stopping early once one scores at least 45, so a class that works on the first seed costs exactly one generation.

A rejected asset never reaches the mixer. Its interval is marked `no_usable_foley` and left silent, the measured values and reason are recorded in the job result, the file is kept on disk for diagnostics, and processing continues for every other action. If every asset is rejected the video is still produced, with a silent track. The mixer holds an independent second limit: no clip is amplified by more than $+25$ dB, and a clip needing more is refused rather than clamped, because clamping still admits amplified noise.

VI. VISUAL EVENT LOCALISATION AND ALIGNMENT

This is the stage that makes the output correct rather than merely present.

Frames are decoded to 320×180 greyscale at 24 fps and motion is computed as the mean absolute inter-frame difference within a horizontal band of the frame. Four bands are defined as fractions of frame height: feet (0.62 – 1.00), head (0.00 – 0.50), table (0.40 – 0.85) and full (0.00 – 1.00). Let $\mathbb{I}_t$ denote frame t restricted to band b ; the motion signal is

$$m_b(t) = (1/|B|) \sum_{p \in B} |I_t(p) - I_{t-1}(p)| \quad (1)$$

smoothed before peak analysis. The band restriction in (1) is what makes the signal usable: measuring motion over the whole frame mixes a footstep into arm and head movement, whereas the feet band responds almost exclusively to gait.

The detection rule depends on the physics of the action, not on the object:

- **footstep** — a prominence peak followed to its subsequent minimum. The swing of the leg is the peak; the *plant* is what is audible.
- **hold** — a sustained motion minimum. A sip is the vessel held still at the lips, so it is a minimum rather than a peak.
- **contact** — the final motion peak before rest, where the object meets the surface as movement stops.
- **continuous** — the whole interval, where no discrete instant exists.

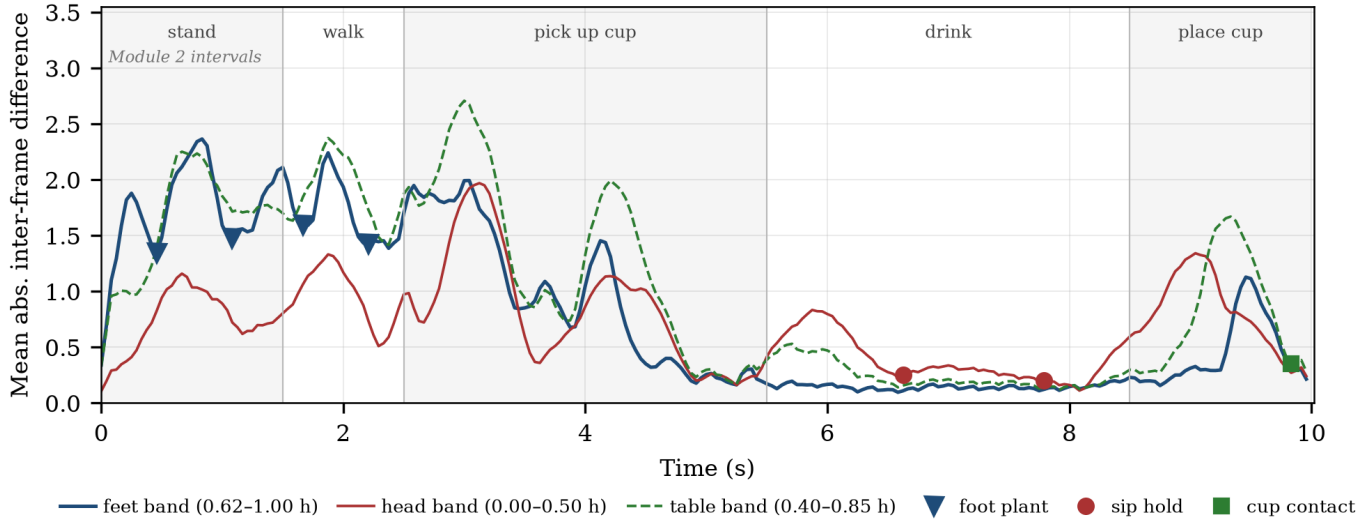


Fig. 2. Region-band motion signals for the reference clip, with the visual events derived from them. Feet-band motion during the interval labelled *stand* (mean 1.708) is statistically indistinguishable from the labelled *walk* interval (1.711), and two of the four foot plants fall inside the *stand* label. Drinking is by a wide margin the least visually active interval, which is why sip holds are detected as motion minima rather than peaks.

A. Alignment uses true attack times

Once a visual instant is known, the generated clip must be positioned so that its audible transient coincides with it. An early implementation aligned onset-*strength* peaks, which is incorrect: an onset-strength peak marks where spectral flux is greatest, not where the transient begins, and it leads or lags the true attack by -96 to $+250$ ms. One asset footstep reported at 3.760 s in fact attacks at 3.856 s, so aligning the strength peak misplaced the audible sound by exactly 96 ms — an error that was audible in the rendered output before it was found.

Alignment therefore uses the true envelope attack: the amplitude envelope maximum, back-tracked to the last point at which it rose through 20 % of that maximum. Clips are shifted, never time-stretched. Where the generated cadence differs from the filmed one the residual is absorbed and reported, because correcting it would alter the character of the audio.

B. Mixing

Per clip, in order: DC removal, cut points snapped to the nearest zero crossing within ± 3 ms, 12 ms raised-cosine fades, level set by active RMS against the class target, and a peak cap at -6 dBFS. Levels differ by class by design: a mug meeting a table is percussive, footsteps are mid-ground, and a sip is intimate, so normalising all three to equal loudness would not correspond to any real recording position. The peak cap is an outlier guard, not a leveller; at -12 dBFS it bound on most clips and flattened the dynamics between events, which is why it was raised.

The bus sums, normalises linearly to -6 dBFS, and passes through a safety limiter (threshold -6 , ceiling -3) whose gain reduction is always reported.

VII. EXPERIMENTAL SETUP

All experiments ran on one machine: Apple M4, 17.18 GB unified memory, macOS 26.2, PyTorch 2.9.1 with MPS. There is no CUDA path. Software versions are pinned per environment as described in Section III.

The reference clip for S2 is a 10.005 s, 1280×720 , 24 fps recording of a person walking to a table, picking up a cup, drinking twice and placing the cup down. It is held under a recorded SHA-256 and was not modified during development. S1 was evaluated on seven recordings of 59–98 frames.

Measurements come from four sources, none of which involved regenerating audio for the purpose of this paper:

- **Job records.** 32 completed pipeline runs persisted to disk, each containing per-event alignment residuals recorded at render time.
- **Generation records.** 38 generation runs, of which 31 used production settings, each recording per-phase timing and memory.
- **The asset corpus.** All 54 audio files produced during development, re-measured with the production validator.
- **Test suites.** 64 automated tests, executed for this paper.

The synthetic vibration experiment for S3 was run specifically for this paper and is described in Section IX-F.

VIII. RESULTS

A. Functional verification

Both automated suites pass in full: 42 tests covering configuration, prompt resolution, video validation, the job state machine, synchronisation, mixing, rendering and the API surface; and 22 tests covering the quality gate and candidate selection. An end-to-end gate over the reference clip completes with five actions detected, four Foley classes generated, zero assets rejected, five placements written to the mix, and one

interval (*stand*) deliberately silent. The gate additionally asserts that no rejected asset reached the mix.

The generalised synchronisation service reproduces the validated reference implementation exactly — foot plants at 0.458, 1.083, 1.667, 2.208 s, sip holds at 6.625 and 7.792 s and cup contact at 9.833 s — and this is asserted in the test suite rather than checked by hand. The band statistics recomputed for Fig. 2 reproduce the recorded values to three decimal places.

TABLE III. SYNCHRONISATION ERROR MEASURED ON THE RENDERED AUDIO OF THE VALIDATED BUILD. ONE FRAME AT 24 FPS IS 41.7 MS.

Action	Visual event	Rendered attack	Error
Walk (plant 1)	0.458 s	0.458 s	−0 ms
Walk (plant 2)	1.083 s	1.063 s	−20 ms
Walk (plant 3)	1.667 s	1.675 s	+8 ms
Walk (plant 4)	2.208 s	2.221 s	+13 ms
Drink (hold 1)	6.625 s	6.638 s	+13 ms
Drink (hold 2)	7.792 s	7.803 s	+11 ms
Place cup	9.833 s	9.833 s	−0 ms

The worst error is 20 ms, below the 20.8 ms half-frame interval at 24 fps. All seven events are within half a frame. We stress that this is one clip with seven events, evaluated on the build that was tuned on it; it demonstrates that the method *can* achieve sub-frame placement, not that it does so in general.

B. Synchronisation accuracy

Table III reports the synchronisation errors for the validated build, measured on the *rendered audio* by detecting envelope attacks in the final WAV and comparing against the visual event timestamps. This is a stronger measurement than reading the alignment plan, because it includes any error introduced by segment selection, fades and mixing.

C. Why alignment error accumulates

The alignment residuals recorded across job runs are deterministic per asset, which permits a more informative analysis than an aggregate. Fig. 3 shows the cause.

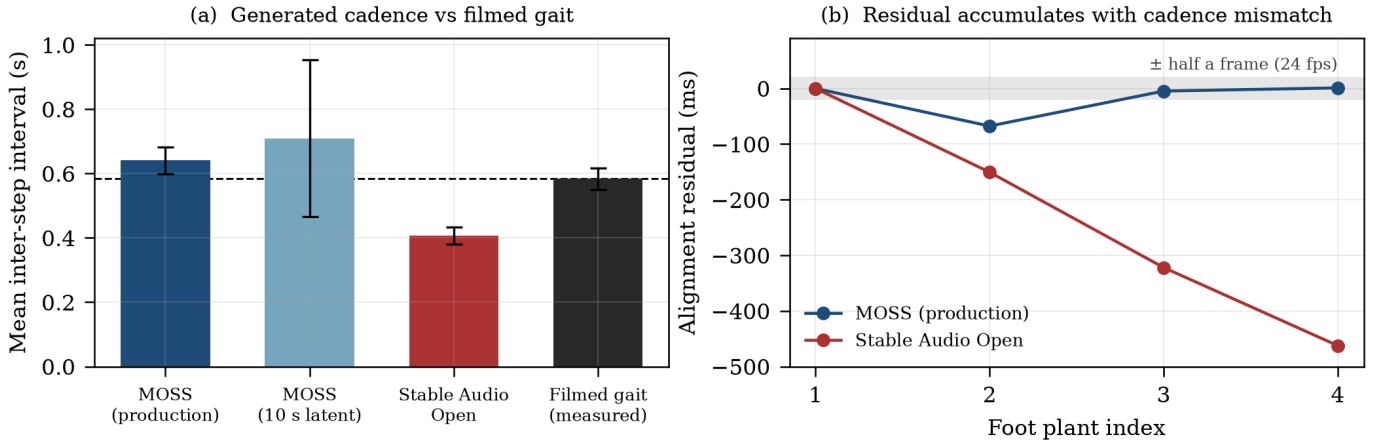


Fig. 3. (a) Mean inter-step interval of each walking asset compared with the filmed gait, with standard deviation. (b) Per-plant alignment residual for the two production backends. Because clips are shifted and never stretched, a mismatch between generated cadence and filmed cadence accumulates across successive contacts.

The filmed gait has a mean inter-step interval of 0.583 s ($\sigma = 0.034$ s). The MOSS walking asset generates steps at 0.641 s ($\sigma = 0.042$ s), 9.9 % slower, and its residuals stay within 0 to −67.6 ms across four plants. The Stable Audio asset generates at 0.407 s ($\sigma = 0.026$ s), 30.2 % faster, and its residual grows monotonically to −462.4 ms by the fourth plant. That growth is predictable from the assets alone: three step intervals at a 0.176 s deficit accumulate to 0.528 s, and 0.462 s was observed after the matcher selected the best-fitting run.

This is a direct consequence of the shift-only policy and is the clearest argument for its limits. Shifting preserves the character of the generated audio but cannot correct a cadence

mismatch, so alignment quality for rhythmic actions depends on the generator producing a plausible tempo. It also gives an objective, timing-based reason to prefer MOSS for this material that is independent of the harmonic-ratio argument in Section III.

D. Quality gate ablation

All 54 assets produced during development were re-measured with the production validator. The gate rejects 16 of them, 29.6 % of the corpus. Fig. 4 shows the corpus in the plane of the two most discriminating metrics.

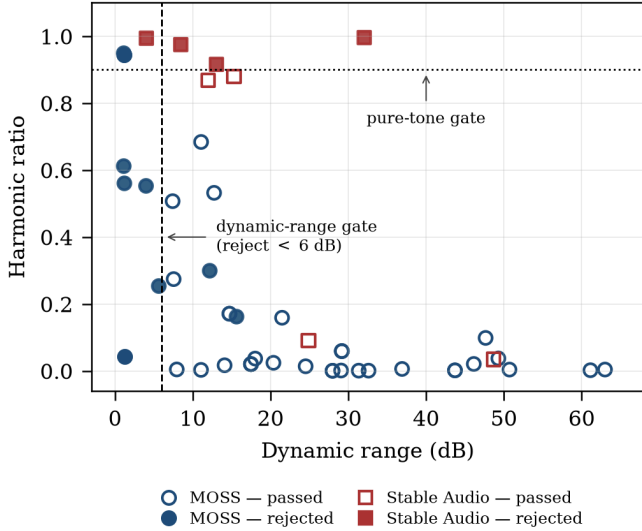


Fig. 4. All 54 generated assets in the dynamic-range / harmonic-ratio plane. Filled markers were rejected. The Stable Audio cluster at harmonic ratio near 1.0 consists of near-pure tones; the two at 32 dB and 12 dB of dynamic range would have passed a dynamic-range test alone.

TABLE IV. QUALITY GATE OUTCOMES BY BACKEND ACROSS THE FULL DEVELOPMENT CORPUS.

Backend	Assets	Rejected	Median score	Median harmonic
MOSS-SoundEffect	46	12 (26 %)	54.5	0.040
Stable Audio Open	8	4 (50 %)	53.2	0.898

The backend split is informative. Median quality scores are almost identical — 54.5 against 53.2 — while median harmonic ratio differs by a factor of 22. The aggregate score does not separate these backends; the harmonic ratio separates them decisively. This is the empirical case for multi-criteria gating: a single scalar, however carefully weighted, can be blind to the failure that matters.

Had the gate not existed, all 16 would have reached the mixer. Eleven of them would have required more than the +25 dB mixer limit, with a median required gain of +29.1 dB and a maximum of +42.1 dB. Applying +42 dB to a file whose peak is −62 dBFS does not produce a quiet contact sound; it produces amplified quantisation noise.

Ten assets have a harmonic ratio above 0.80. The two most extreme, both Stable Audio cup placements, measure 0.997 and 0.995 with spectral flatness of 0.00000 — pure sine waves. Two assets at 0.880 and 0.868 pass, which indicates the gate has a real boundary rather than rejecting everything tonal.

E. Generation settings: what the metrics miss

Two speed-ups were tried during development and reverted after listening. Table V quantifies both.

TABLE V. GENERATION-SETTINGS ABLATION. THE AGGREGATE SCORE IS INSENSITIVE TO THE STEP REDUCTION AND MOVES IN A MISLEADING DIRECTION FOR THE LATENT REDUCTION, WHILE THE STRUCTURAL MEASURE DETECTS THE DEGRADATION.

Setting	Gen (s)	Score	Dyn (dB)	Attacks	Gap σ
<i>(a) Denoising steps — impact class</i>					
50 steps (production)	266	86.9	32.6	7	—
35 steps	200	86.3	31.3	7	—
25 steps	129	86.4	29.1	7	—
<i>(b) Denoised latent length — walking class</i>					
30 s latent (production)	242	97.1	43.7	16	0.042
10 s latent	88	86.0	61.1	8	0.244
Filmed gait (reference)	—	—	—	4	0.034

Reducing the step count from 50 to 25 halves generation time and moves the quality score by 0.6 points, within noise. The transient count is unchanged. The gate is effectively blind to this change, and the decision to revert it rests entirely on listening.

The latent reduction is more revealing. Cutting the denoised window from 30 s to 10 s makes generation 2.8×

faster and *raises* both the measured dynamic range (43.7 to 61.1 dB) and the effective bits (14.9 to 16.0) — both of which the score rewards. Yet the number of detectable transients falls from 16 to 8, and the standard deviation of inter-step intervals rises from 0.042 s to 0.244 s, a factor of 5.8.

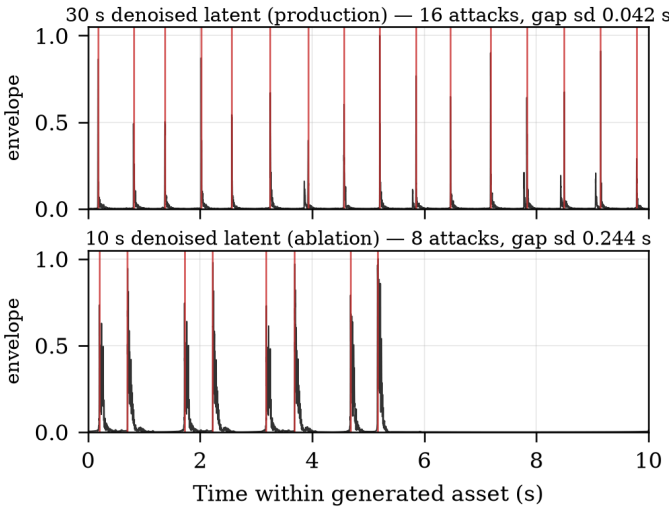


Fig. 5. Amplitude envelope of the walking asset generated with the production 30 s latent (top) and the 10 s ablation (bottom). Vertical lines mark detected attacks. The ablated asset has half as many transients and 5.8 $\times$ the inter-step irregularity, while scoring higher on dynamic range.

The decisive case. Cup pickup is the class for which multi-candidate generation exists. With the 30 s latent, one of three seeds passes the gate (seed 44, score 86.6). With the 10 s latent, **none** of the three passes; peaks sit at -33 dBFS and all three would need over $+40$ dB of gain. The speed-up would have made the hardest class permanently silent.

The general lesson is that an aggregate score computed from marginal statistics can improve while the temporal structure that makes audio usable is destroyed. A structural measure — how many transients are present and how regularly they are spaced — captures what the scalar misses, and is cheap to compute.

F. Acoustic Eye: controlled characterisation

The vibration path had no positive result on real footage, because the available recordings are at 30–60 fps and the recoverable band is bounded by half the frame rate. To establish whether the implementation is correct, a synthetic stimulus with known ground truth was constructed: a band-limited noise texture translated sinusoidally by a known sub-pixel amplitude at a known frequency, rendered at several frame rates. This is an optical test, not an acoustic capture.

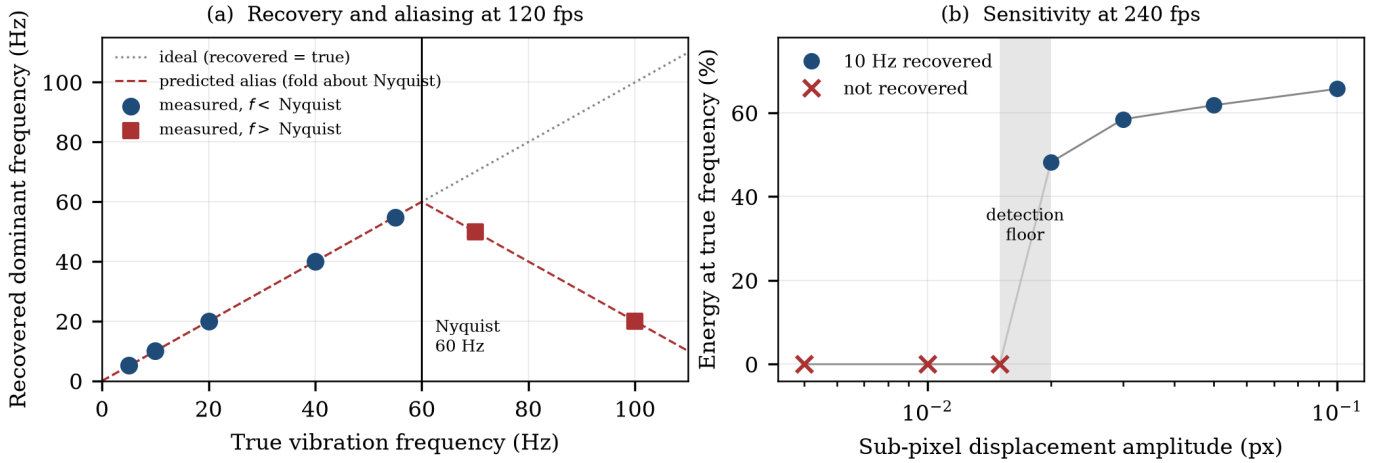


Fig. 6. Controlled characterisation of the visual microphone on synthetic stimuli. (a) At 120 fps, frequencies below Nyquist are recovered exactly, and frequencies above it fold about Nyquist exactly as the dashed prediction requires. (b) Recovery at 240 fps degrades sharply between 0.020 px and 0.015 px of displacement.

The results are unambiguous. A 10 Hz oscillation of 0.30 px amplitude is recovered with zero frequency error at every frame rate from 30 to 480 fps, carrying 65–66 % of the spectral energy above 0.5 Hz. At 120 fps, signals at 5, 10, 20, 40 and 55 Hz are recovered exactly; signals at 70 and 100 Hz fold to 50 and 20 Hz respectively, matching the aliasing prediction to within measurement resolution. Sensitivity degrades sharply below 0.02 px: recovery is exact at 0.020 px with 48.2 % of the energy at the true frequency, and fails completely at 0.015 px.

The implementation is therefore correct, and its limit on ordinary footage is the sampling theorem rather than a defect. At 60 fps the recoverable band ends at 30 Hz, which excludes speech and essentially all musical content. The recorded 60 fps job produced a signal whose dominant component is at 0.06

Hz, which is camera drift, not sound. Reporting the system as a demonstrator of the principle, subject to a stated frame-rate requirement, is the only honest description.

Throughput was measured on the same machine: 1284 frames/s at 64×64 , 441 frames/s at 128×128 , and 103 frames/s at 256×256 . Cost scales with pixel count, as one steerable pyramid is constructed per frame.

G. Latency and resource use

TABLE VI. MEASURED LATENCY AND MEMORY. S1 AND S2 FIGURES ARE MEANS OVER 7 AND 31 RECORDED RUNS RESPECTIVELY; S3 IS MEASURED ON SYNTHETIC INPUT.

Path	Quantity	Measured
S1	Model load	0.5–0.7 s
S1	Inference (87 frames mean)	1.57 s
S1	Total per clip	3.73 s
S1	Real-time factor	0.45×
S2	Phase 1 — text encoder	4.7 s
S2	Phase 2 — diffusion (50 steps)	264.4 s
S2	Phase 3 — decode	8.6 s
S2	Total per generated asset	280.9 s
S2	Peak resident memory	12.11 GB
S2	Minimum available memory	2.35 GB (worst 1.58)
S2	Whole job, fully cached	39.5 s (median)
S2	Whole job, with generation	556.9 s (median)
S3	Throughput, 128×128	441 frames/s

S1 runs faster than real time on CPU. S2 is dominated by diffusion, which consumes 94 % of generation time; because the model denoises a fixed-length latent, a short request costs the same as a long one. Caching matters accordingly: a fully cached job completes in about 40 s against roughly nine minutes when generation is required.

The memory figures deserve attention. Peak resident memory averaged 12.11 GB against a 17.18 GB machine, and in the worst recorded run the available memory fell to 1.58 GB against a guard threshold of 1.50 GB — a margin of 80 MB. The memory guard was never breached across 31 recorded generations, and swap growth was 0.00–0.01 GB. The phase separation described in Section IV-E is what makes this fit; without it the three components would be co-resident and the run would fail.

H. Audio independence of the lip-reading path

A claim that a system reads lips must be demonstrated rather than asserted. The audio stream was stripped from a test recording with `ffmpeg -map 0:v:0 -c:v copy -an` and the clip re-run. Decoded video frames were bit-identical to the original, with a maximum pixel difference of 0 across 2.21 billion values; the model input tensor was bitwise identical; and the output matched exactly, with the same sentence, token identifiers and decoder score. The result is structural as well as empirical: the model frontend is a 3-D convolutional stack accepting only (B, T, 1, 88, 88) pixel tensors, and cannot ingest a waveform.

I. Input format sensitivity in S1

TABLE VII. S1 DECODER SCORE AGAINST INPUT FORMAT. HIGHER (LESS NEGATIVE) IS MORE CONFIDENT.

Format	Input tensor mean	Best score
25 fps SDR	0.06	−0.36
30 fps SDR	0.57–0.61	−1.05
30 fps HDR (HLG)	1.24–1.29	−4.31

Auto-AVSR is trained on 25 fps SDR video, and HDR footage decoded without tone mapping shifts the pixel distribution far from the training regime. The system standardises every upload to 25 fps SDR and rejects HDR rather than converting it, because a correct HLG-to-SDR conversion requires an FFmpeg filter that is absent from many builds, and silently degrading accuracy is worse than refusing the input. We note that these are decoder scores, not accuracy: they indicate the model’s confidence, and a confident wrong answer is possible.

IX. LIMITATIONS

Action recognition is the binding constraint on S2. Foley can only be as good as the timeline it is given. On a coffee-stirring test video, the recogniser missed a cup placement entirely, emitted one stirring action under three different labels, and returned a caption (“Stirring a cup of coffee”) rather than an action. Fig. 2 shows a second instance: the interval labelled *stand* has feet-band motion of 1.708 against 1.711 in the labelled *walk*, and contains two of the four foot plants. The pipeline compensates by widening the footstep search span, but the label itself is wrong.

Synchronisation is validated on limited footage. The 20 ms result is seven events on one clip, on the build tuned against it. Across the 32 recorded job runs, only footstep placements record per-event residuals; hold and contact placements record the aligned instant but not a residual, so the multi-clip evidence is thinner than the number of runs suggests.

No word error rate is reported for S1. Ground-truth transcripts for the test recordings were not retained, so recognition quality is characterised only by decoder scores and by two spot-checked correct transcriptions. The published checkpoint reports 20.3 % WER on LRS3; no claim is made that this system reproduces that figure on its own material.

Listening evaluation is single-assessor. Every judgement that audio “sounds right” was made by the developer. No formal listening test with multiple participants, randomised presentation or statistical analysis was conducted. Given that Section VIII-D demonstrates objective metrics failing to capture perceptible degradation, this is a substantive limitation rather than a formality.

S3 has not recovered real sound. The characterisation in Section VIII-F is synthetic. No real acoustic signal has been recovered from real footage, because no high-speed camera was available.

Output is sparse by nature. For discrete object interactions, roughly 18 % of a 10 s timeline carries audio. The silence between events is correct, but it means the result is not a continuous soundtrack.

Apple Silicon only, single machine, single job. The pipeline depends on MPS with no CUDA or CPU fallback. Jobs run in process with no queue, authentication or multi-user isolation. Video length is capped at 60 s because recognition cost grows linearly.

Comparative evaluation is limited. No comparison against Diff-Foley, MMAudio or FoleyCrafter on a standard benchmark with standard metrics was performed. The

rejections in Section III are measurements on this project's own material and hardware, and should not be read as general statements about those systems.

X. FUTURE WORK

The most valuable next step is improving the action timeline, since it bounds everything downstream in S2. Two directions are concrete: prompt engineering for the vision-language model to elicit action phrases rather than captions, and label normalisation to merge synonymous predictions across windows.

A second direction is a formal listening evaluation with multiple assessors, randomised presentation and a pairwise design. The finding that scores are insensitive to audible degradation makes a perceptual study the only way to validate the quality gate's thresholds.

Third, the shift-only alignment policy could be relaxed for rhythmic actions. Section VIII-C shows that residual accumulation is driven by cadence mismatch; selecting among several generated candidates by cadence fit, rather than by quality score alone, would address this without time-stretching.

Fourth, the visual microphone path should be evaluated on genuine high-speed footage. The synthetic characterisation establishes the implementation is correct; a 240 fps capture of a loud source next to a light, resonant object is the natural test.

Finally, a comparison against Diff-Foley, MMAudio and FoleyCrafter on VGGSound or a comparable benchmark, reporting Fr chet Audio Distance alongside onset accuracy, would place the alignment method in context.

XI. CONCLUSION

This paper described a system that reconstructs audio from silent video along three paths — visual speech recognition with timed synthesis, action recognition with aligned Foley generation, and phase-based vibration recovery — running entirely on a single consumer laptop.

The central finding is that in generated-Foley systems, placement and validation matter as much as generation. Anchoring generated audio to visually measured contact instants using true envelope attack times achieved a worst-case error of 20 ms on the reference clip, within half a frame at 24 fps. Measuring every generated asset before mixing rejected 29.6 % of the development corpus, eleven items of which would otherwise have received more than +25 dB of automatic gain.

Two results argue against trusting aggregate audio quality scores. Two backends with nearly identical median scores, 54.5 and 53.2, differ in median harmonic ratio by a factor of 22, one of them producing pure sine waves for object contacts. And a generation setting that raised measured dynamic range from 43.7 to 61.1 dB simultaneously halved the number of detectable transients and made a class that previously produced one usable candidate in three produce none. In both cases a structural measure caught what the scalar missed.

The system is a demonstrator with real limits, and they have been stated: action recognition is the weakest component, synchronisation is validated on limited footage, the listening evaluation has one assessor, and the vibration path has been characterised only on synthetic stimuli. Within those limits, the work shows that a laptop-scale pipeline can place generated sound accurately enough to survive frame-level scrutiny, provided the audio it generates is measured before it is believed.

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